GradientShop

Differentiable Discrete-Event Shopfloor World Models as Sandboxes for Manufacturing Agents

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Project duration:	12 months (10/2026 – 9/2027)
Requested amount:	95.000 €

Attachments, to be appended to this form (please submit the entire application as a single PDF!):

1. CV with list of publications (max. 2 pages)
2. Statement by the supervisor confirming
    a. support for the planned project,
    b. the scientific independence of the project, including independent management of the budget, and
    c. funding for the applicant's position for the duration of the project.

Abstract:

Discrete-Event Simulation (DES) is a core planning tool in production engineering for scheduling, intralogistics and energy management on shopfloor. Industry is now actively integrating Large Language Models (LLMs) into existing engineering software like DES. However, DES is event-driven and non-differentiable, while LLMs are continuous and gradient-based. Rare mathematical framework yet couples them at the representation level. GradientShop builds the experimental and methodological foundation to bridge this gap, developing a differentiable DES as an intrinsic world model for LLM agents to enable rapid closed-loop learning and establishing a benchmark on a battery remanufacturing process. The results seed follow-on DFG and ERC proposals on agentic factory operations.

1. Project description *(max. 6000 characters)*

Please describe your project (topic, research questions, objectives, current state of research and your own preliminary work, relevance to research/society). Please also explain how the proposed project differs from your previous activities and projects. You may structure your text with subheadings.

Topic and motivation

The global push to integrate AI into manufacturing, reflected in national programs across the EU [1], the US [2], and China [3], is driving a broad effort to embed Large Language Model (LLM) Agents (LLMA) into established industrial software. Germany has long led in this domain, with PLM and CAx developed by companies, such as Siemens AG, forming the backbone of global manufacturing engineering. **Discrete-Event Simulation (DES)** is a core planning tool within this ecosystem, and the field is now embracing the integration of LLMAs into established DES [4]. Unlike software engineering, where a compiler gives coding agents a cost-free feedback loop, every shopfloor trial costs material, downtime and safety. A **DES world model (WM)** can substitute, but a regular DES still forces random sampling, which is too slow for practical use. Making DES differentiable provides gradient feedback directly to the agentic LLM, making this coupling a necessity.

State of research and gap

The core scientific obstacle is **the absence of a mathematical framework that couples DES and LLM due to the incompatibility of their two underlying representations.** LLMs operate over continuous token embeddings trained by gradient descent, whereas DES executes branching, event-driven logic with non-differentiable state transitions that expose no gradient to an external learner.

Manufacturing research treats the simulator as a black box behind an API. LLMA observes only end-of-run scalars, with no ability to replay or rewind internal states. Every policy improvement reexecutes the simulation, which is sample-inefficient and renders trial-and-error practically infeasible. Through APIs, LLMs were used to generate DES models [5], to drive scheduling [6], to optimize warehouse slotting [7] and to save energy [8]. Even in the wider field, only preliminary deeper couplings exist, with ReflectiChain embedding a supply-chain WM into the inference loop [9] and SGA pairing an LLM with a differentiable simulation in molecular design [10]. No existing work embeds DES into LLMA.

The AI community pursues WMs and physicsAI mainly in the visual domain (NVIDIA Cosmos [11] and Isaac GR00T [12]), where physics is grounded in pixel space and gradient flow is natural when coupled with LLMs. On the other hand, **Differentiable DES (DDES)** is established through smoothing branch conditions [13], and continuous relaxation was scaled to queueing-network control with 50–1000× sample-efficiency gains [14]. Gumbel-Softmax [15] and Monte Carlo estimators [16] support gradient estimation through discrete variables. All of these advances originate in generic settings, none in manufacturing shopfloor or integrated in LLMA.

Research questions

GradientShop aims to establish the foundation of a DDES Shopfloor WM as a sandbox for manufacturing LLMA (Fig. 1).

RQ1: Can continuous relaxation of DES preserve sufficient structural fidelity under the combinatorial routing and stochastic processing times of real shopfloor environments?

RQ2: Can the forward replay of a DDES be embedded as a structured reasoning step within LLM inference, grounding linguistic self-evaluation in simulation-backed state trajectories?

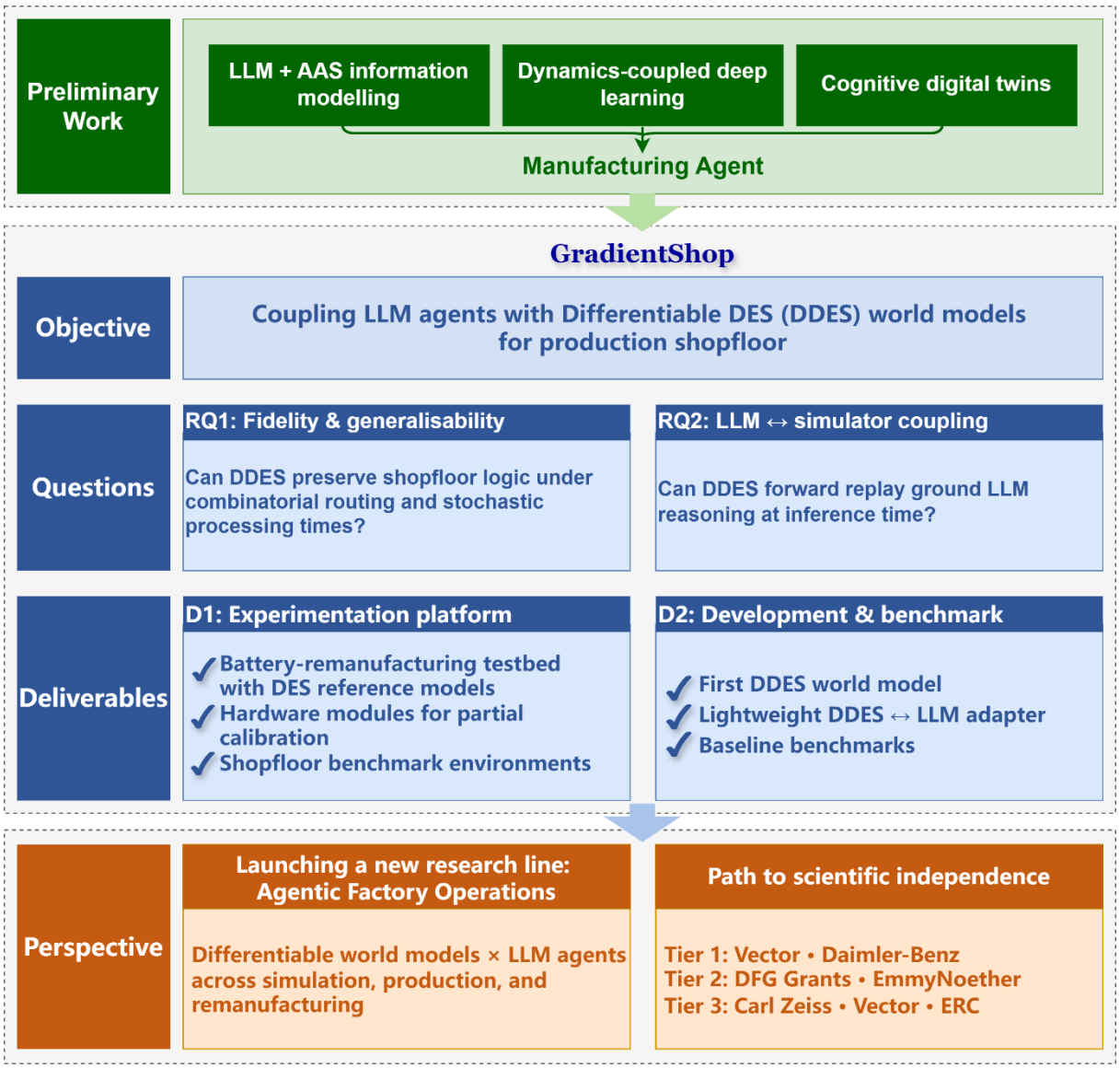


Fig. 1 Overall research plan

Objectives and deliverables

The proposed project will lay the essential experimental and methodological groundwork for addressing the two RQs in a rigorous and systematic manner. By generating key preliminary results and validating the underlying research framework, the project will create a strong foundation for future DFG and ERC proposals aimed at advancing these questions in greater depth. Two deliverables serve this purpose.

D1 is an integrated **shopfloor experimentation platform** in a battery remanufacturing scenario, including DES reference models, physical hardware modules, and benchmark task formulation.

- The software reference comprises a commercial DES model (Siemens Tecnomatix Plant Simulation or AnyLogic) as an industry-standard reference, and a parallel open-source model (SimPy/salabim) that exposes full event-driven semantics for differentiability research.
- The hardware modules cover the battery testing stage and intralogistics, using existing IFF infrastructure. They supply calibration data for DES models and are extendable for future plant-wide use.
- A benchmark suite of shopfloor tasks (scheduling [6], warehouse slotting [7], energy optimization [8]) implemented on the DES models forms the basis for all numerical experiments.

D2 is the exploratory development and experiments that build on the D1 platform, comprising a first differentiable DES, a lightweight LLM coupling adapter, and benchmark experiments on both RQs.

- A first DDES is implemented, applying continuous relaxation techniques to discrete dispatch decisions and smoothed approximations of event-time dynamics. This probes RQ1 by identifying where DDES preserves fidelity, where it breaks down, and which open problems should anchor follow-on proposals.
- A lightweight adapter couples this DDES with an agentic LLM as a proof of concept toward RQ2. The LLM proposes a dispatch policy, a forward DDES replay returns gradients on the relevant parameters, and this signal grounds the LLM's next token prediction.
- Benchmark experiments establish quantitative baselines against the DDES-WM and define the comparison point for future work.

Relevance/Impact

The platform's scientific value lies in the benchmark and reference infrastructure that the community currently lacks. It lowers the barrier for follow-on research on PhysicsAI.

DES is widely employed in logistics, healthcare operations, transportation, communication networks and energy systems. A DDES-WM-LLMA coupling mechanism transfers directly to these domains and multiplies the long-term impact.

Preliminary work and differences from previous activities

The applicant's doctoral and postdoctoral research produced a manufacturing LLMA based on cognitive Digital Twins (DT) in which Asset Administration Shell carry semantic data and simulations serve as the DT physics engine callable via APIs. This LLMA will be used in D2, requiring no further agent development.

These results gave me command of LLMA and simulation, yet none addressed DES, scheduling, or differentiable simulation. The methodological core of GradientShop (DDES and the LLM coupling adapter) is therefore new and constitutes my independent research identity.

GradientShop is equally distinct from the host institute. Prof. Bauernhansl leads IFF and IPA on Industrie 4.0, cyber-physical production, automation and robotics, and biointelligent manufacturing, none of which targets the GradientShop core. IFF supplies the production-engineering context, industry network and remanufacturing knowledge, while the methodological core is mine to establish.

2. Perspective *(max. 1500 characters)*

Please describe the contribution to the applicant's independence and autonomy. Explain why funding from Boost Your Science! during this period would be particularly beneficial for your research. Also explain how prospective external funding (third-party funding provider/funding format/timeline) is planned.

As shown in Fig. 2, my **long-term vision** is to establish a **theoretical foundation for differentiable engineering WMs with integrated LLMA** as a novel *modelling method* that fuses symbolic, neural and physics representations across the *asset lifecycle* for AFO (independent group to be established).

The vision spans three axes. The *digital integration* axis defines how a factory is brought under agent control, the *asset lifecycle* axis what is covered, and the *modelling method* axis how it is modelled by fusing physics, neural and symbolic representations into one differentiable WM.

Prior LLMA work mainly sits on the first axis (TRL 5). **GradientShop is the entry point for the modelling method axis** as the true methodological core, starting with DDES-WM in the production phase and extensible to CAx WMs later. Establishing this core requires independent ownership of budget and topic, which is what BYS provides.

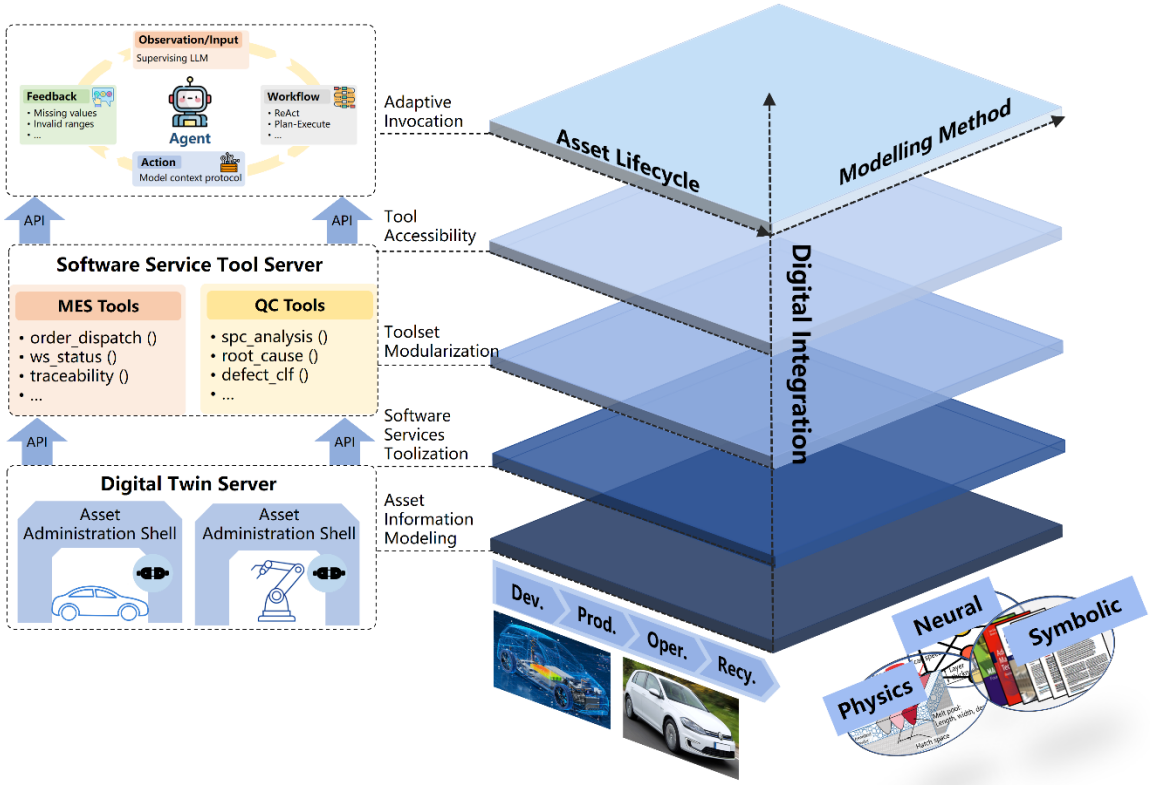


Fig. 2 Conceptualized research line: Agentic Factory Operations (AFO)

Timing is decisive. Career-wise, four postdoctoral years have built the expertise and visibility for 1.5M grant bids. BYS is the first step to independence and the platform for them. Scientifically, the 2025 LLMA wave triggered a WM and physicsAI surge from 2026, yet none have reached production engineering. This is the moment to claim a position before the field converges.

External funding is staged in three tiers. **Tier 1** runs companion projects alongside BYS that will be applied in parallel with GradientShop and strengthen the GradientShop kernel. **Tier 2** turns the work into an independent line and secures continuation of my contract. **Tier 3** consolidates everything into flagship grants. The specific instruments, timeline and content are given in §3.1.

3. Work programme, requested funds, perspective

3.1 Work programme and schedule

Describe the work steps relevant to the project in text and in a table, including the planned schedule for implementation. (max. 2000 characters)

The 12-month project (Oct 2026 – Sep 2027) is organized in three task groups.

Task 1 (M1–M5) builds the experimentation platform.

Subtask	Description	Output
1.1	SHK staffing, environment and the commercial DES license are set up first	Production environment
1.2	build reference DES models for remanufacturing process; collect data	Reference models
1.3	align reference DES models, hardware status and benchmark scope	Calibrated reference models
1.4	develop a benchmark suite from the validated platform on selected shopfloor tasks	Benchmark suite

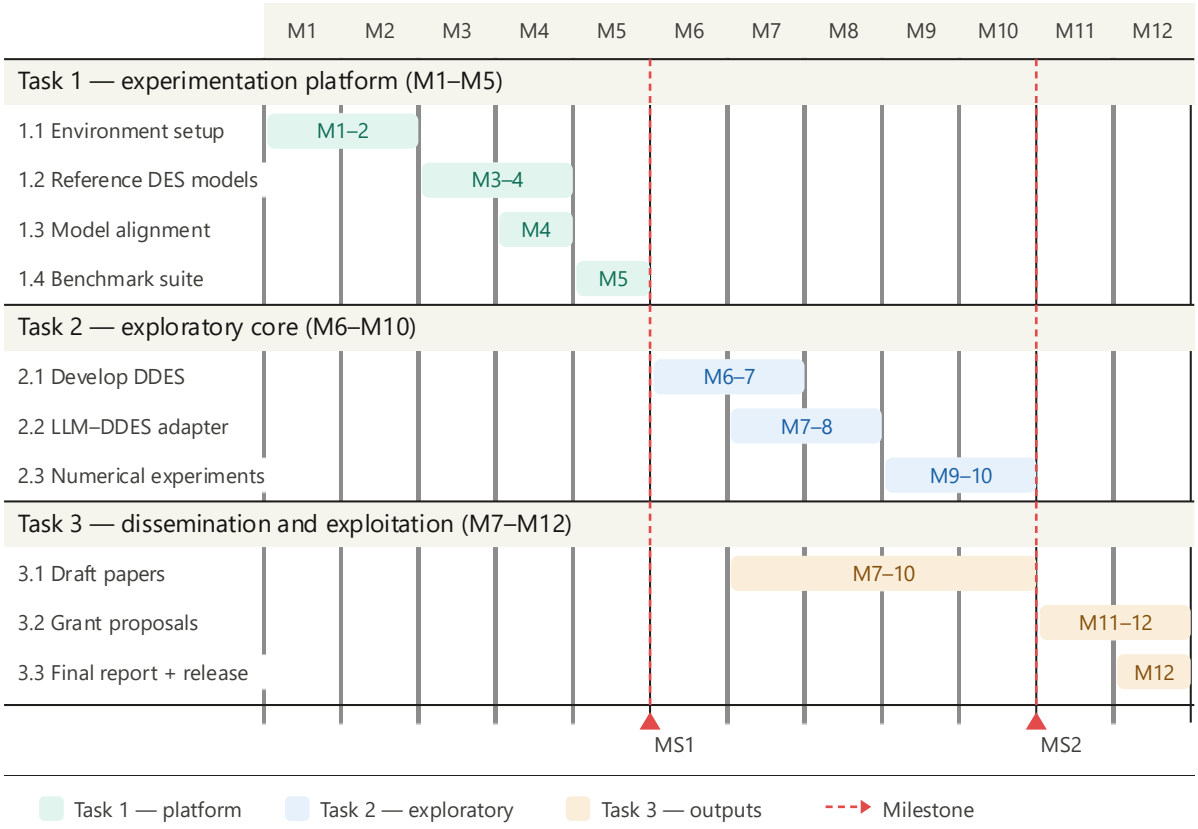
Task 2 (M6–M10) is the exploratory core, incl. developing DDES and DDES-WM-LLMA and experimental benchmark.

Subtask	Description	Output
2.1	develop initial DDES	DDES algorithms
2.2	train adapter for LLM-DDES integration	DDES-WM-LLM
2.3	conduct numerical experiments; refine DDES-WM-LLM and the benchmark suite	Experiment results

Task 3 (M7–M12) deals with the scientific dissemination and exploitation, especially for future grant applications.

Subtask	Description	Output
3.1	draft papers for two conferences and Journal of Manufacturing Systems;	Submitted Papers
3.2	**draft proposal for Tier-1 and Tier-2 grants**	Submitted Proposal
3.3	write final report; release GradientShop v1.0 + benchmark suite	Final report and open-source software package

The **schedule** is as follows:



Task 3.2 aims to prepare applications for the following **external funding** sources at Tier 1 and Tier 2. Tier 3 grants are planed to consolidate the previous proposal and project results.

Grants	Tier 1	Tier 2	Tier 3
Objective	Parallel companion for BYS	Independence & continuation	Flagship consolidation
Instruments	**D**aimler-**B**enz **S**tiftung; **V**ector **S**tiftung; ZIM (partner found)	**D**FG **E**mmy **N**oether; **D**FG **I**ndiv. **G**rant; EU/German coop. projects	**C**arl-**Z**eiss-**S**tiftung; **V**ector Junior Group Leader; **ERC** Start Grant
Timeline	DBS & VS Oct. 2026; ZIM Sep. 2026	DEN Nov 2026; DIG 2027 at BYS mid-term; Coop. projects ongoing	Submit end-2027 / 2028
Role vs. Gradient-Shop & AFO	**DBS**: exploring alternative DDES formulations and gradient estimators	**DEN**: proposal of the entire AFO vision (4y PostPhD)	If DEN bid is unsuccessful, **three 1.5M grants** will be applied in parallel, built on an updated AFO vision according to GradientShop results and state of research
	VS: exploring alternative DDES-WS-LLM fusion	**DIG**: derived from open questions in GradientShop	
	ZIM: empower a wood-processing machine with an LLMA function	**EU Horizon / Federal Ministry** Coop. projects to secure contract continuity	

3.2 Requested funds

3.2.1 Budget plan

Please enter the project costs in the following table.

What? (personnel and material resources, travel, etc.)	Cost
Personnel — SHK, commercial DES implementation, 40 h/mo × 12 mo	11,000 €
Personnel — SHK, open-source DES implementation, 40 h/mo × 12 mo	11,000 €
Personnel — SHK, hardware integration of the testing module, 40 h/mo × 12 mo	11,000 €
Material — Commercial DES software license, academic seat	6,000 €
Material — Commerical LLM API, GPU server usage fee	16,000 €
Investment — BaSyTec CTS 8-channel cell cycler (Germany), incl. software licence	18,000 €
Investment — Cell holders, cylindrical CBH-8 plus pouch plus coin, with Kelvin cabling	3,000 €
Investment — Type-90 LIB safety storage cabinet, EN 14470-1, TRGS 510 compliant	6,000 €
Material — EoL automotive cells from Duesenfeld or KIT Battery LabFactory, cables, fuses, BMS dev-board, PPE	3,000 €
Material — electrolyte, separator, electrode foils, coin-cell hardware (CR2032/pouch), glove-bag supplies	3,000 €
Material — reference electrodes, EC-Lab/BaSyTest licence extensions, calibration standards	3,000 €
Travel — 2 international conferences	4,000 €
Total cost	**€ 95,000**

3.2.2 Explanation of the budget plan

Please explain the costs listed under 2.2.1. If requested funds are also to be used by others the requested funds must be particularly justified. (max. 500 characters)

Personnel are three student assistants covering commercial DES, open-source DES and hardware integration respectively. Hardware spend is intentionally limited to the battery testing module. Other functions leverage existing IFF infrastructure or will be covered by other grants. The commercial DES license, LLM API and GPU server usage fee serve the PI's research only. No item is shared with third parties.

References

Please list the references mentioned in the project description.

[1] European Commission, Apply AI Strategy, "AI for Manufacturing, Engineering and Construction — flagship actions under the Apply AI Strategy" (2025), 8 October 2025

[2] Lear Corporation / Palantir Technologies. Five-year partnership expansion press release, 4 Sep 2025. lear.com/newsroom.

[3] CPC Central Committee. Suggestions for the 15th Five-Year Plan (2026–2030), adopted Oct 2025 (4th Plenary Session, 20th CC).

[4] Siemens Digital Industries Software. Plant Simulation User Conference 2025 recap, Jul 2025. blogs.sw.siemens.com/tecnomatix.

[5] Schmitt, T. et al. LLM-driven discrete-event simulation: a generative AI framework for automated model generation, adaptation, and evaluation in manufacturing. Journal of Manufacturing Systems 85:642–661, 2026.

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[9] Luo, J., From topology to trajectory: LLM-driven world models for supply chain resilience. arXiv:2604.11041, 2026.

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[13] Andelfinger, P. Differentiable agent-based simulation. ACM SIGSIM-PADS, 2021.

[14] Che, E., Dong, J. & Namkoong, H. Differentiable discrete-event simulation for queuing network control. arXiv:2409.03740, 2024.

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